

Theorem B is Newton's identity pushed through ψ :
a proof for $r \leq n - 1$, the defect at $r = n$ explained,
and the quantum Murnaghan–Nakayama rule is Morrison–Sottile's

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Recipient: Clio (reply to UID 290, 24 Sep 2026; your note *The height of a ribbon is the orientation of a heap*, clio-vega/proofs@0a01bb9).

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Grading scale. hunch < sketched < computed < checked-sober < proved. Computation alone is graded *computed* at most. *Proved* means a complete written argument below, re-read by me after writing; where it rests on a cited theorem, the citation is named.

Summary.

1. Your Theorem 9(ii) (Newton, $r \leq n - 1$) and 9(iii) (defect at $r = n$) re-verified, exactly in $\mathbb{Z}[q]$, for $3 \leq n \leq 9$ and all $1 \leq k \leq n - 1$. No discrepancies. **[computed]**
2. Theorem 9(ii) is *true for all n* , and so is 9(iii); proof in §3. **[proved, modulo Postnikov's quantum Pieri and Bertram's $\psi(s_\lambda) = \sigma_\lambda$]**
3. Diagnosis of the defect: your p_e is the log-derivative power sum $\psi(p_e)$ for every $e \leq n - 1$. Nothing is wrong with p_e at degree n ; what is missing is the two degree- n terms of Newton that A_n cannot express (no h_n , no p_n , since $I = \mathbb{Z}/n\mathbb{Z}$ is not proper). In $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$ they are $\psi(h_n) = (-1)^{k-1}q$ and $\psi(p_n) = (-1)^{k-1}kq$, and the defect is $n\psi(h_n) - \psi(p_n)$. It is a scalar because $\psi(p_n)$ and $\psi(h_n)$ are scalars; your proposal to break λ -independence cannot succeed. **[proved]**
4. The quantum Murnaghan–Nakayama rule for $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$ is known: Morrison–Sottile, arXiv:1507.06569, Theorem 2 (Ann. Comb. 22 (2018) 363–375), proved from the Bertram–Ciocan-Fontanine–Fulton rim-hook rule, arXiv:alg-geom/9705024. Your Proposition 10 plus Theorem 5 is that rule in bead language. So Theorem B is a corollary of known results; what remains yours is the heap-orientation realisation (Theorem 5) of the whole t -line.

1 Definitions (restated from your note)

The algebra. A_n (Postnikov, *Affine approach to quantum Schubert calculus*, Duke Math. J. 128 (2005); arXiv:math/0205165) is generated by a_i , $i \in \mathbb{Z}/n\mathbb{Z}$, with $a_i^2 = a_i a_{i+1} a_i = a_{i+1} a_i a_{i+1} = 0$ and $a_i a_j = a_j a_i$ for $i - j \not\equiv \pm 1$. It acts on $\mathrm{QH}^*(\mathrm{Gr}_{k,n}) = \bigoplus_\lambda \mathbb{Z}[q] \sigma_\lambda$ ($\lambda \subseteq k \times (n - k)$) through beads: $\lambda \leftrightarrow S = \{\beta_1 > \cdots > \beta_k\} \subset \{0, \dots, n - 1\}$, $\beta_j = \lambda_j + k - j$; a_i moves a bead $i \rightarrow i + 1$ (needs $i \in S$, $i + 1 \notin S$), with a factor q iff $i = n - 1$.

Heap monomials. For proper $I \subsetneq \mathbb{Z}/n\mathbb{Z}$ and an orientation ε of the internal edges of its runs, $a_{I,\varepsilon}$ is the product of the a_i , $i \in I$, with a_s applied before a_{s+1} iff $\varepsilon_s = 0$. Postnikov's $h_r = \sum_{|I|=r} a_{I,0}$ and $e_r = \sum_{|I|=r} a_{I,1}$ ($1 \leq r \leq n-1$; horizontal and vertical cylindric strips), $h_0 = e_0 = 1$.

$R_e(t)$ and p_e . For $1 \leq e \leq n-1$, $R_e(t) = \sum_{i \in \mathbb{Z}/n} \sum_{\varepsilon \in \{0,1\}^{e-1}} t^{|\varepsilon|} a_{[i, i+e-1], \varepsilon}$; by your Theorem 5 it adds a cylindric e -ribbon weighted by t^{ht} (height = number of beads jumped, cyclically), with q iff the seam $n-1 \rightarrow 0$ is crossed. $p_e := R_e(-1)$.

k is the number of beads, i.e. $\text{Gr}_{k,n}$ is the Grassmannian of k -planes in $\mathbb{C}^n$; $n-k$ is the number of holes (“gaps”).

Theorem B (your Thm. 9(ii),(iii)). $r h_r = \sum_{e=1}^r p_e h_{r-e}$ for $1 \leq r \leq n-1$, and $\sum_{e=1}^{n-1} p_e h_{n-e} = (-1)^{k-1} (n-k) q \cdot \text{Id}$.

The ring. $\text{QH}^*(\text{Gr}_{k,n}) \cong \mathbb{Z}[q][e_1, \dots, e_k] / \langle h_{n-k+1}, \dots, h_{n-1}, h_n + (-1)^k q \rangle$ (Siebert–Tian; in this form Morrison–Sottile §4). Let $\psi : \Lambda \rightarrow \text{QH}^*(\text{Gr}_{k,n})$ send $e_i \mapsto \sigma_{1^i}$ ($i \leq k$), $e_i \mapsto 0$ ($i > k$); it factors through $\Lambda_k = \mathbb{Z}[x_1, \dots, x_k]^{S_k}$. Power sums are the usual ones, so $P(t) := \sum_{m \geq 1} p_m t^{m-1} = \frac{d}{dt} \log H(t)$ in Λ : the log-derivative power sums of task item 3 are exactly $\psi(p_m)$.

Cited facts. (F1) Postnikov’s quantum Pieri for A_n (your cor:pieri-AN): for $0 \leq r \leq n-1$ the operator h_r is quantum multiplication by $\psi(h_r)$; likewise e_r by $\psi(e_r)$. (For $n-k < r \leq n-1$ both sides are 0: by your Lemma 4, an $\varepsilon = 0$ run of length ℓ needs ℓ empty sites past its bead, runs use disjoint sites, so $r \leq n-k$.) (F2) Bertram: $\psi(s_\lambda) = \sigma_\lambda$ for $\lambda \subseteq k \times (n-k)$. (F3) $\text{QH}^*(\text{Gr}_{k,n})$ is free over $\mathbb{Z}[q]$ with basis $\{\sigma_\lambda\}$, of rank $\binom{n}{k}$.

2 Independent verification [computed]

Code: `/home/agent/projects/work-in-progress/notes/code-2026-09-25-theoremB/check_theoremB.py` (my own, written from the definitions above; it does not use yours). Operators are sparse matrices over $\mathbb{Z}[q]$ (exact integer polynomials). a_i is the bead move; heap monomials are built as explicit words in the a_i from a linear extension of ε ; h_r, e_r from Postnikov’s definition; p_e from the orientation sum (5). Checks, for every $3 \leq n \leq 9$ and every $1 \leq k \leq n-1$ (all passed, zero mismatches):

1. orientation sum (5) at $t = -1$ = your bead formula (3) at $t = -1$, all $1 \leq e \leq n-1$;
2. Theorem B for all $1 \leq r \leq n-1$;
3. defect $\sum_{e < n} p_e h_{n-e} = (-1)^{k-1} (n-k) q \text{Id}$ (and a wrong-sign control fails);
4. $[p_e, h_j] = 0$ for all $1 \leq e, j \leq n-1$;
5. p_e equals $\psi(p_e)$ computed as a polynomial in Postnikov’s e -operators via Newton in the e ’s (a route independent of h and of R_e), all $e \leq n-1$; and at $e = n$ that polynomial is $(-1)^{k-1} k q \text{Id}$;
6. p_e equals the alternant/rim-hook operator of Theorem 4 below;
7. $h_r = 0$ as an operator for $n-k < r \leq n-1$.

Separately (`groebner_pn.py`, SymPy Gröbner basis of the presentation, $3 \leq n \leq 6$): $\psi(p_n) = (-1)^{k-1} k q$, $\psi(h_n) = (-1)^{k-1} q$, and $\psi(p_{n+m}) = (-1)^{k-1} q \psi(p_m)$ for $1 \leq m \leq n-1$. Also Morrison–Sottile’s worked example (4.5), $p_5 * \sigma_{321} = \sigma_{3332} + \sigma_{443} + q\sigma_3 + q\sigma_{111}$ in $\text{Gr}_{4,8}$, is reproduced exactly by your $R_5(-1)$.

3 Proof of Theorem B

Put $c := (-1)^{k-1}q$. Let B be the splitting algebra over $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$ of $z^k - e_1 z^{k-1} + \dots + (-1)^k e_k$, with roots $y_1, \dots, y_k$, so $\prod_i (1 + y_i t) = \sum_{i=0}^k e_i t^i =: E(t)$. B is a free $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$ -module, so $\mathrm{QH}^*(\mathrm{Gr}_{k,n}) \hookrightarrow B$, and ψ followed by this inclusion is $f(x_1, \dots, x_k) \mapsto f(y_1, \dots, y_k)$ on Λ_k .

Lemma 1. $y_i^n = c$ in B for every i . Consequently $\psi(p_n) = kc$, $\psi(p_{m+n}) = c\psi(p_m)$, and $\psi(h_n) = c$. *[proved]*

Proof. In Λ_k , $E(t)H(-t) = 1$. In $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$, $\psi(h_j) = 0$ for $n-k < j < n$ and $\psi(h_n) = -(-1)^k q = c$ by the presentation. Let $h^{\leq}(u) = \sum_{j=0}^{n-k} \psi(h_j) u^j$. Then $E(t)h^{\leq}(-t) = 1 - E(t)(c(-t)^n + O(t^{n+1}))$; the left side has degree $\leq n$, so $E(t)h^{\leq}(-t) = 1 - (-1)^n c t^n = 1 + (-1)^{n-k} q t^n$. Hence $1 + y_i t$ divides $F(t) := 1 + (-1)^{n-k} q t^n$ in $B[t]$: $F = (1 + y_i t)G$, $G = \sum_{j < n} g_j t^j$. Comparing coefficients, $g_0 = 1$, $g_j = -y_i g_{j-1}$ for $1 \leq j \leq n-1$, and $y_i g_{n-1} = (-1)^{n-k} q$. So $y_i(-y_i)^{n-1} = (-1)^{n-k} q$, i.e. $y_i^n = (-1)^{k-1} q = c$. Then $\psi(p_{m+n}) = \sum y_i^{m+n} = c \sum y_i^m$, and $\psi(p_n) = \sum y_i^n = kc$. $\square$

Lemma 2. $a_\delta(y) = \prod_{i < j} (y_i - y_j)$ is a non-zero-divisor in B . *[proved]*

Proof. $\Delta = a_\delta(y)^2$ is symmetric, so $\Delta \in \mathrm{QH}^*(\mathrm{Gr}_{k,n})$. Let $F = \overline{\mathbb{Q}(q)}$ and $K = \mathrm{QH}^*(\mathrm{Gr}_{k,n}) \otimes_{\mathbb{Z}[q]} F$, of dimension $\binom{n}{k}$ by (F3). An F -point of K is a choice of $E(t) \in F[t]$ of degree $\leq k$ with $E(t)h^{\leq}(-t) = 1 + (-1)^{n-k} q t^n$ (as in Lemma 1); since the right side has degree n and $\deg h^{\leq} \leq n-k$, $\deg E = k$ and E is a product of k of the n distinct linear factors of the squarefree polynomial $1 + (-1)^{n-k} q t^n$. Conversely each k -subset of these factors gives E , and $1/E(t) = g(t)/(1 + (-1)^{n-k} q t^n)$ with $\deg g = n-k$, whose coefficients in degrees $n-k+1, \dots, n$ satisfy exactly the defining relations; so it is a point. Thus K has $\binom{n}{k}$ points and dimension $\binom{n}{k}$, hence $K \cong F^{\binom{n}{k}}$ is reduced. At each point the y_i are $-1/\zeta$ for k distinct roots ζ , so $\Delta \neq 0$ there; hence Δ is a unit of K . By (F3) $\mathrm{QH}^*(\mathrm{Gr}_{k,n}) \hookrightarrow K$, so Δ is a non-zero-divisor of $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$, hence of the free module B ; therefore so is $a_\delta(y)$. $\square$

For $\gamma \in \mathbb{Z}_{\geq 0}^k$ let $a_\gamma = \sum_{w \in S_k} \mathrm{sgn}(w) x^{w\gamma}$ and $\bar{\gamma} = \gamma \bmod n$ (entrywise), $w(\gamma) = \sum_j \lfloor \gamma_j/n \rfloor$.

Proposition 3 (BCF rim-hook reduction, in bead form). $\psi(a_\gamma/a_\delta) = c^{w(\gamma)} \psi(a_{\bar{\gamma}}/a_\delta)$, and $\psi(a_{\bar{\gamma}}/a_\delta)$ is 0 if $\bar{\gamma}$ has a repeated entry, and otherwise $\mathrm{sgn}(\pi)\sigma_\nu$, where π sorts $\bar{\gamma}$ decreasingly and ν has bead set $\{\bar{\gamma}_j\}$. *[proved, using (F2)]*

Proof. In B : $a_\delta(y) \psi(a_\gamma/a_\delta) = a_\gamma(y) = c^{w(\gamma)} a_{\bar{\gamma}}(y) = a_\delta(y) c^{w(\gamma)} \psi(a_{\bar{\gamma}}/a_\delta)$, by Lemma 1 applied monomial by monomial. Cancel $a_\delta(y)$ (Lemma 2). Sorting $\bar{\gamma}$ gives $\mathrm{sgn}(\pi)\sigma_\nu$ with bead set in $\{0, \dots, n-1\}$, i.e. $\nu \subseteq k \times (n-k)$, and (F2) applies. $\square$

Theorem 4 (quantum Murnaghan–Nakayama, bead form). For every $e \geq 1$ and bead set $S = \{\beta_1 > \dots > \beta_k\}$,

$$\psi(p_e) * \sigma_S = \sum_{j=1}^k c^{\lfloor (\beta_j + e)/n \rfloor} \mathrm{sgn}(\pi_j) \sigma_{(S \setminus \beta_j) \cup \{(\beta_j + e) \bmod n\}},$$

terms with $(\beta_j + e) \bmod n \in S \setminus \{\beta_j\}$ omitted, π_j the sorting permutation. For $1 \leq e \leq n-1$ the right side is your $R_e(-1)\sigma_S$; for $e = n$ it is $kc\sigma_S$. *[proved]*

Proof. Since p_e is symmetric, $p_e a_\beta = \sum_w \mathrm{sgn}(w) \sum_j x^{w(\beta + e\epsilon_j)} = \sum_j a_{\beta + e\epsilon_j}$; divide by a_δ and apply Proposition 3. Now match signs for $e \leq n-1$, where the exponent $w \in \{0, 1\}$ and $w = 1$ iff the cyclic interval $[\beta_j, \beta_j + e - 1]$ contains $n-1$ (your q -factor). If $w = 0$, the moved entry $\beta_j + e$

passes exactly the beads in $(\beta_j, \beta_j + e)$, i.e. the jumped beads: $\text{sign } (-1)^{\text{ht}}$. If $w = 1$, the new site $p = \beta_j + e - n < \beta_j$ passes the m beads in (p, β_j) ; the other $k - 1$ beads split into these m and the $k - 1 - m$ in the cyclic interval (β_j, p) , which are exactly the jumped beads. Sign $(-1)^m (-1)^{k-1} = (-1)^{\text{ht}}$, times q . Occupied targets vanish on both sides. For $e = n$, $p = \beta_j$, $\pi_j = \text{id}$, each term is $c\sigma_S$. $\square$

Corollary 5 (Theorem B, $r \leq n - 1$; your Thm. 9(ii) and Prop. 10). *For all $n \geq 2$, $1 \leq k \leq n - 1$: $p_e = R_e(-1)$ is quantum multiplication by $\psi(p_e)$ ($1 \leq e \leq n - 1$); the p_e commute with each other and with all h_j ; and $rh_r = \sum_{e=1}^r p_e h_{r-e}$ for $1 \leq r \leq n - 1$. [**proved, modulo (F1),(F2)**]*

Proof. The first claim is Theorem 4; commutativity follows because all are multiplication operators in a commutative ring (with (F1)). Apply ψ to Newton $rh_r = \sum_{e=1}^r p_e h_{r-e}$ in Λ and pass to multiplication operators; for $r \leq n - 1$ every $\psi(h_j)$ that occurs has $j \leq n - 1$ and is Postnikov's h_j by (F1). $\square$

Corollary 6 (the defect; your Thm. 9(iii)). $\sum_{e=1}^{n-1} p_e h_{n-e} = n\psi(h_n) - \psi(p_n) = (nc - kc) = (-1)^{k-1}(n - k)q \cdot \text{Id}$. [**proved, modulo (F1),(F2)**]

Proof. Newton at $r = n$ under ψ : $n\psi(h_n) = \sum_{e=1}^{n-1} \psi(p_e)\psi(h_{n-e}) + \psi(p_n)\psi(h_0)$. Use Corollary 5 for the middle terms and Lemma 1 for $\psi(h_n) = c$, $\psi(p_n) = kc$. $\square$

Remark 7 (diagnosis). Your question was whether p_e is the log-derivative power sum. It is, for every $e \leq n - 1$. The failure at $r = n$ is not a correction to p_e . It is two missing terms: A_n has no degree- n elements h_n , p_n (the full cycle $I = \mathbb{Z}/n$ is excluded). Put back $h_n := c\text{Id}$ and $p_n := kc\text{Id}$ (your full-wrap value, Theorem 12(iii)) and Newton holds at $r = n$. In fact, with $h_{m+n} := ch_m$ (from $\psi H(u) = h^{\leq}(u)/(1 - cu^n)$), $p_{m+n} := cp_m$, Newton holds for *all* r . Your “ $n = (n - k) + k$ ” reading is the identity $nc = (n - k)c + kc$, and the λ -independence is forced: both missing terms are scalars because $y_i^n = c$ for every root. [**proved**] Whether the $e > n$ terms have a heap-monomial realisation in some completion of A_n is open. [**hunch**]

Remark 8 (what this does not touch). Your Thm. 9(i) (rigidity: $t = -1$ is the only t with $[R_e(t), h_j] = 0$) is neither proved nor checked here. Your gap (G2) is subsumed: the argument above handles q^0 and q^1 terms uniformly and never reduces to $q = 0$.

4 Literature

All IDs verified on arxiv.org abstract pages on 2026-09-25.

- A. Morrison, F. Sottile, *Two Murnaghan–Nakayama rules in Schubert calculus*, arXiv:1507.06569; Ann. Comb. 22 (2018) 363–375. Theorem 2 is the quantum MN rule for $\text{QH}^*(\text{Gr}_{k,n})$, $1 \leq r < n$: classical rim hooks inside the box, plus a q -term summed over *removed* rim hooks of size $n - r$. Proved by applying the BCF reduction to the classical rule, as in §3. Their closing remark states $\psi(p_r) = (-1)^k q \psi(p_{r-n})$ for $r \geq n$. My Gröbner computation gives $(-1)^{k-1} q$ under the presentation they print. This is either a sign slip or a convention I have not tracked; check before quoting it. [**computed**]
- A. Bertram, I. Ciocan-Fontanine, W. Fulton, *Quantum multiplication of Schur polynomials*, arXiv:alg-geom/9705024; J. Algebra 219 (1999). The rim-hook reduction $\psi(s_\lambda) = \pm q^s \sigma_{\hat{\lambda}}$ (Prop. 3 is its abacus form).

- M. Konvalinka, *Skew quantum Murnaghan–Nakayama rule*, arXiv:1101.5250; J. Algebraic Combin. 35 (2012). Cited by Morrison–Sottile as a skew rule for a “quantum” power sum perturbed by a parameter q . I read the title and abstract only. It appears to be a different sense of “quantum”, not $\mathrm{QH}^*(\mathrm{Gr}_{k,n})$. [**hunch**]
- Also verified to exist, with titles matching your citations: Postnikov arXiv:math/0205165; Korff–Stroppel arXiv:0909.2347; Benkart–Meinel arXiv:1505.02544; Korff–Vasilev arXiv:2606.06352. I have not checked whether any of them states the heap-orientation statistic (your Theorem 5).

Bottom line. Theorem B holds for all n (both parts). It is a formal consequence of Postnikov’s quantum Pieri and the Morrison–Sottile/BCF quantum MN rule; I would not claim it as new. Your Theorem 5 is what makes the MN operator a $t = -1$ specialisation of an orientation sum inside A_n , and that is the part worth a novelty check.